Florida Cyber Security Clinic

Team 1

CIS 4914 Senior Project

Sasha Watanabe, Team Lead

Yuliang Huang, Business Analyst

Maxwell Griffin, Scrum Master

Adriel Barzola, Backend Development

Leo Graham, Frontend Development

Joseph N. Wilson, Ph.D.

jnw@ufl.edu

Final Presentation: https://youtu.be/R22r_MyqrSc

Keywords: cyber security, clinic, vulnerability assessment

Project Overview

Name: Florida Cyber Security Clinic

Description: Cybercriminals have increased their attacks against all types of targets. Many organizations, especially in local government, critical infrastructure, or nonprofits, do not have the resources to appropriately secure their digital infrastructure. As a result, these organizations are often prime targets for cyber attacks, which, if successful, can compromise the well-being of the people they serve. We present a student-led cybersecurity clinic to help local government, nonprofits, and potentially operators of critical infrastructure secure their organizations by applying existing defensive methodologies. The clinic will conduct a non-intrusive assessment of an organization's security defenses and incident response procedures.

Goals and objectives

- Establish a procedure for ensuring student volunteers are qualified.
- Establish procedures for clinic engagements with clients.
- Develop a platform for managing tasks during clinic engagements.

In-scope items

- Create a prototype portal where clients can send files and make tickets
- Write a general questionnaire for the client to fill out before the first meeting with the client
- Create an online exam for prospective student volunteers
- Write vulnerability assessment criteria related to cyber incident risk and incident response procedures

- Conduct a vulnerability assessment for a client

Out-of-scope items

- Online portal ready for clinic and client use during engagements
- Tailored questionnaires for each type of client
- Vulnerability assessment criteria related to AI security

Roles and responsibilities

Adriel Barzola: Back End Developer

- Designs and implements the functionality of our product. This entails analyzing the client's environment and generating actionable recommendations and advice to address security blind spots and improve its security posture.

Leo Graham: Front End Developer

- Develops and refines the front-end of our product as a means of enhancing the user experience for clients.

Maxwell Griffin: Scrum Master

- Organizes major sections of the project into manageable tasks spread out over a reasonable schedule for the team to follow.

Sasha Watanabe: Team Lead

- Oversees the project’s main objectives, directs the team’s focus toward the more critical tasks first, and changes direction of the project in the face of developments that may change the scope of the project.

Yuliang Huang: Business Analyst

- Point of contact with the client and is responsible for relaying information to and from the team and the client.

Deliverables

Functional exam environment for proctors, students.

- Our website contains a proctor-generable exam that tests the aptitude of students that are planning on working for the clinic.
- This exam pulls from a bank of relevant questions, and is held behind credentials generated by a proctor (whose exam creation dashboard is also held behind credentials.)

Ticket creation, management system.

- A ticket management system is present within the website that allows clinic participants to accept created tickets and resolve them.

File upload system.

- Client environment where files can be uploaded to the clinic.

Assessment criteria.

- Development and integration of criteria for the vulnerability assessment of the client.

Communication templates.

- Several templates for clinic-client communication to standardize the process of initial contact, including questionnaire and client agreement templates.

Vulnerability assessment.

- A draft + finalized vulnerability assessment of our client encompassing all communications and material transfers between the clinic and the client.

Technical Details

Codebase overview (repositories, branches, version control)

The repository is on GitHub.

Key dependencies and libraries

Docker

Deployment instructions (environment details, configuration)

Change the secret key in line 31 of website/docker/website/webapp/app.py.

Database schema and data migration details

Software architecture

Sections of the web application are isolated from each other to minimize unnecessary access. The application consists of multiple Docker containers, each comprising one aspect of our web application: the client area and the student exam (Appendix A). The database is also isolated from the web server when possible in order to mitigate the possible effects of attacks.

Testing Information

Testing summary

We use Python's mypy library (leveraging static type hints) to conduct static code analysis to check for common coding issues. This ensures code quality and catches many common errors before runtime and reduces the amount of bugs that get pushed into production code. We conducted static application security testing using Pyflakes, Bandit, and SonarQube. Pyflakes and Bandit were run through a program called prospector to test for errors and security issues in Python code. We used SonarQube for VSCode to look for issues in non-Python code.

Issue log

Issue	Planned mitigation	Progress updates	Status
401 Unauthorized message when attempting to navigate to login page	Changed the condition for checking if the user is currently logged in from `flask_login.user_logged_in` to `flask_login.current_user.is_authenticated`.	(4/14) Fix tested locally (4/21) Fix pushed to repository	Complete
Underline under ticket message field made it difficult to read the text in the client section	Added `class="ticket_message"` to the abbr tag in the html.	(4/21) Fix tested locally (4/21) Fix pushed to repository	Complete
Parts of navigation bar have inconsistent height when page width is condensed	Edit CSS for text-overflow with a hover effect (show entire text on hover)	(4/21) Fix tested locally	In-progress

User Documentation

Quick Start

First, you'll need to fetch the application from GitHub. You can do this by installing Git for your system, then running

```
git clone https://github.com/UFSecurityClinicSpring2026/website.git
```

To deploy this app, first install Docker for your system. If you're on Windows or you have privileges to run Docker as the current user, you can omit *sudo* from the following commands.

Once you have Docker installed, simply run the following command to get it up and running:

```
sudo docker compose up
```

If you need to stop the Docker containers, run

```
sudo docker compose down
```

If changes are made to the app, you'll need to rebuild it. Run the following command before bringing the app up in order to rebuild it:

```
sudo docker compose build
```

Note that rebuilding the app may clear all data previously associated with the app.

If deploying for production, please change all the secrets in **docker-compose.yml** away from the defaults before deploying the app.

To access the website, open your browser and go to **https://localhost**.



Installing Git

Windows:

1. Go to https://gitforwindows.org/ and click Download.
2. Run the downloaded installer and follow the instructions. In most cases, the default options are okay.
3. Restart your computer.

Debian-based Linux:

1. Run `sudo apt install git`.

Installing Docker

Windows

1. Visit https://www.docker.com/products/docker-desktop/.
2. Follow the installation instructions.

Debian-based Linux

1. Follow the instructions at https://docs.docker.com/engine/install/debian/.

Student Exam

For Proctors

The exam website is located in the subdirectory **/exam**.

The default proctor username is **proctor** and the password is specified in an environment variable in **/docker-compose.yml**.

Each exam is 20 questions long, and students have 1 hour to complete it.

Exam-related data is stored in the **/examdata** directory:

- **questions.csv** - A CSV file containing the question bank for the exam.
- **examstorage.enc** - The encrypted database information.

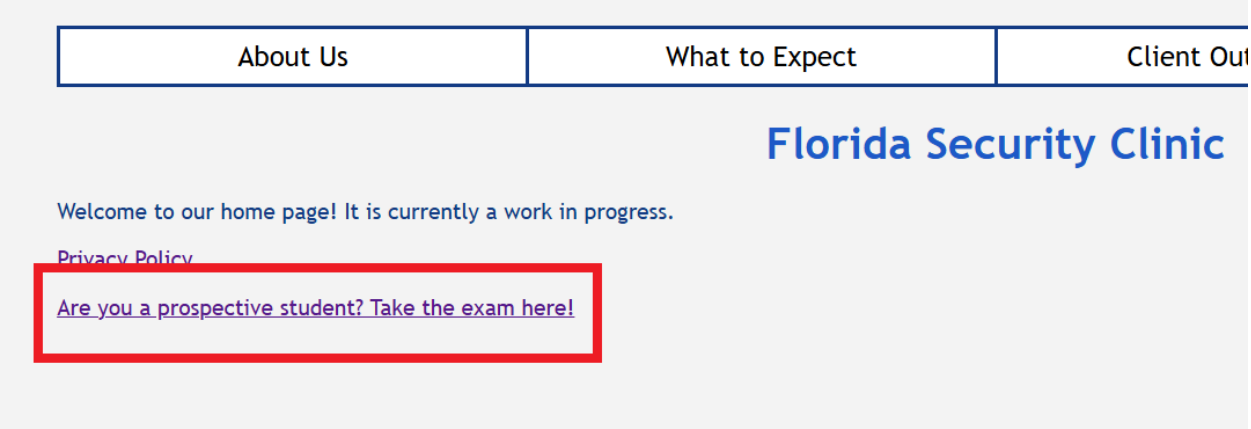
Limitation: The exam database is limited to a size of a little under 256 MiB.

Environment Variables:

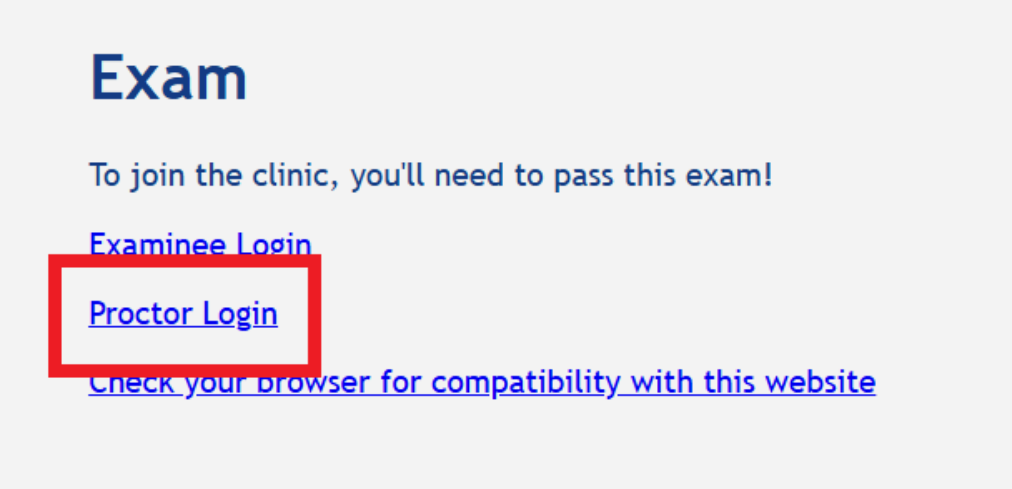
- **DEFAULT_PROCTOR_PASSWORD** - The initial password for the proctor interface login. Please change this password before deploying
- **FILE_ENCRYPTION_SECRET** - The secret for the encrypted files. You should change this to a long, unique password before first deployment. It is unlikely that this password will need to be manually typed over the lifetime of this app. Do not change this value after deploying the app, otherwise data loss may occur. (Not implemented yet)

To create a login for a student to take the exam:

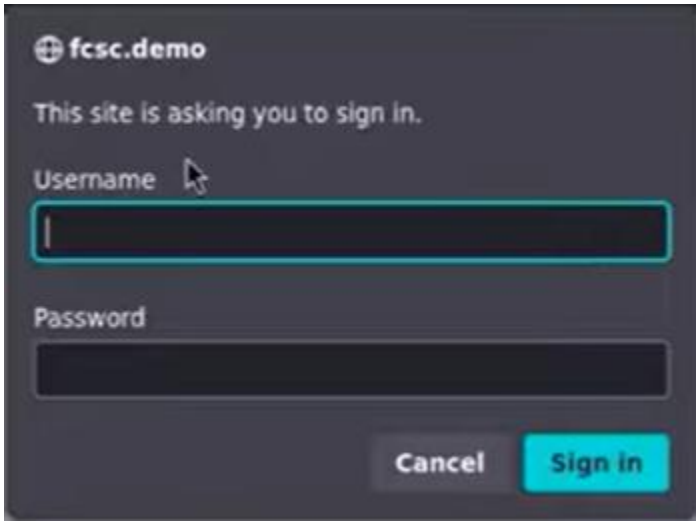
On the website homepage, click **Are you a prospective student? Take the exam here!**



Click **Proctor Login**.



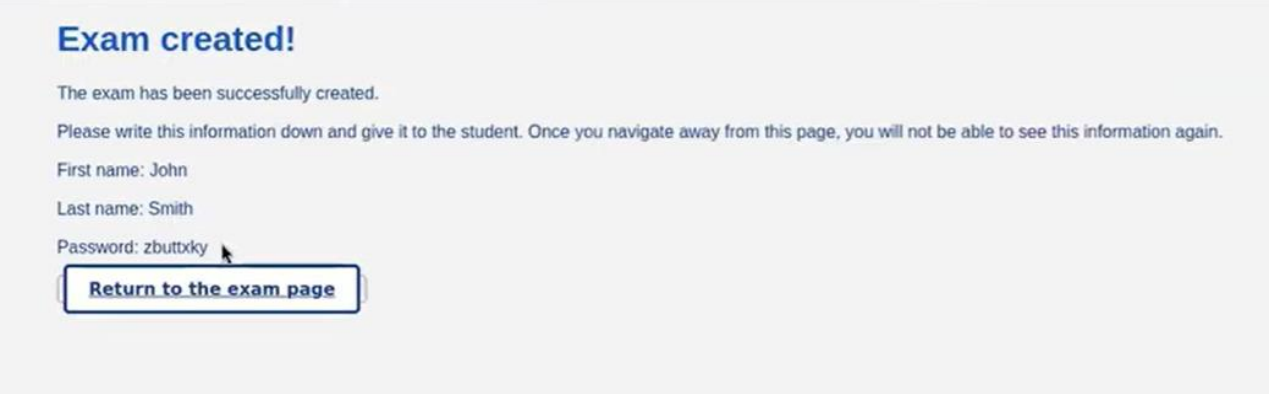
Sign in with the proctor username and password.



Enter the student's first and last name, then click **Create exam**.

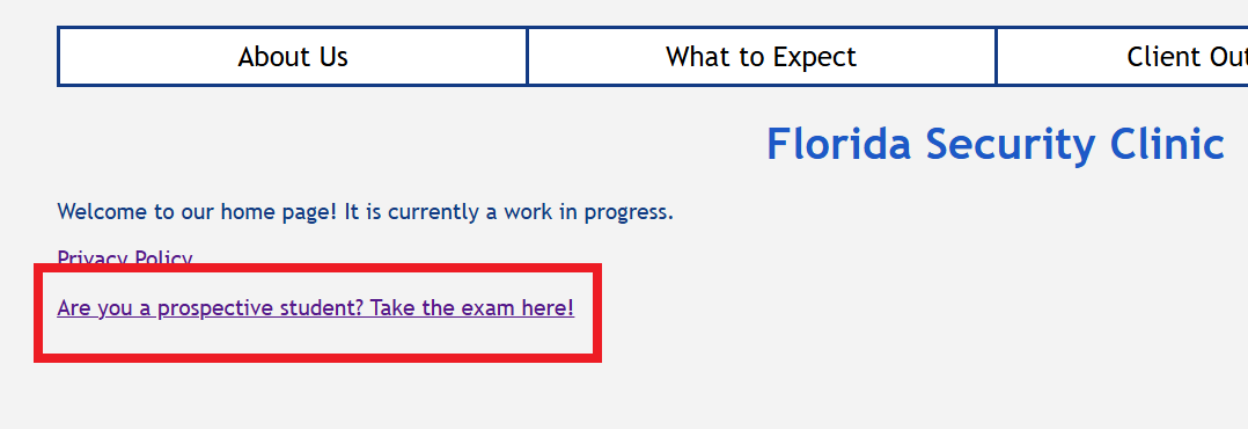


Note down the first and last name and generated password displayed on the screen. Provide the information to the student. To finish, click **Return to the exam page**.

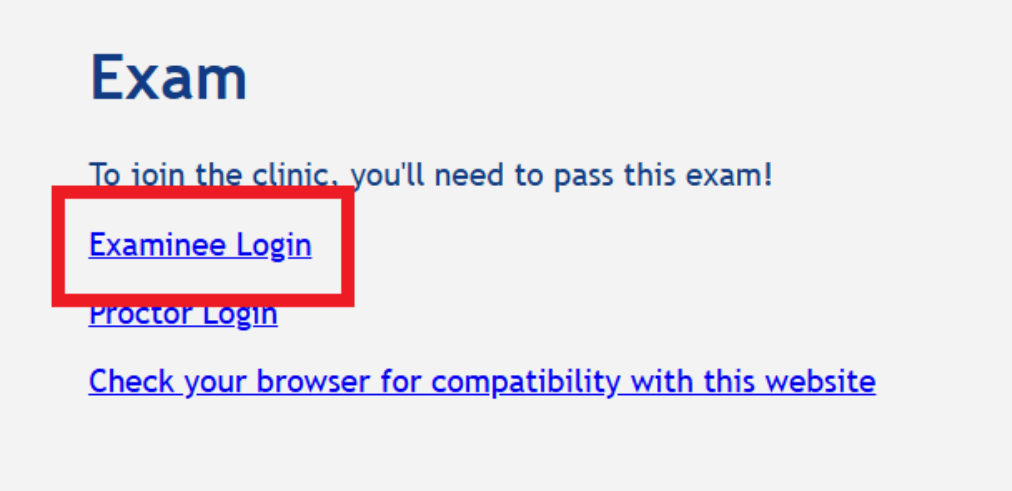


For Students

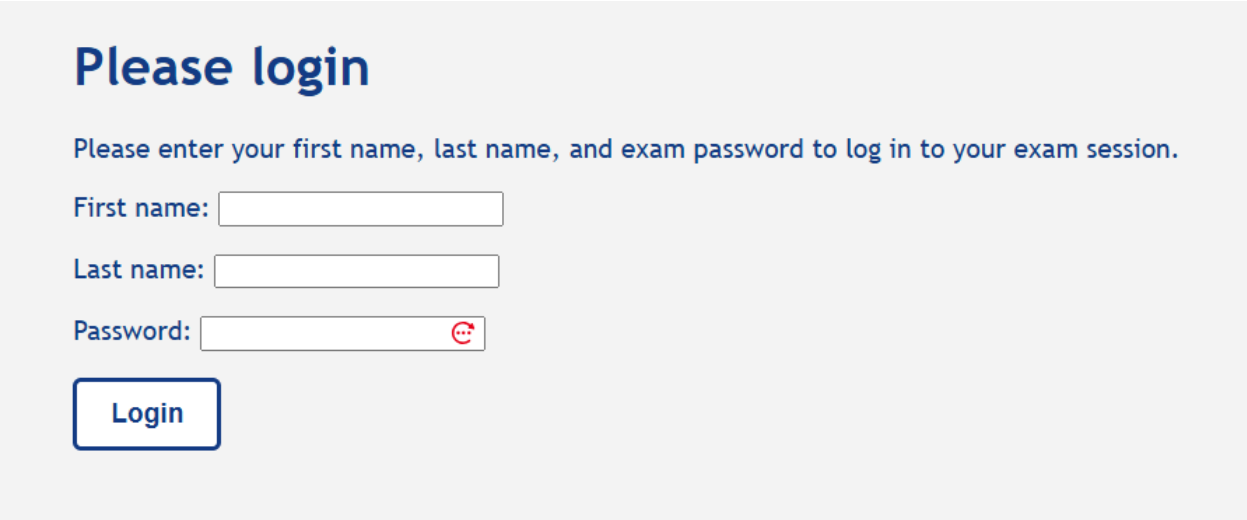
On the website homepage, click **Are you a prospective student? Take the exam here!**



Click **Examinee Login**.

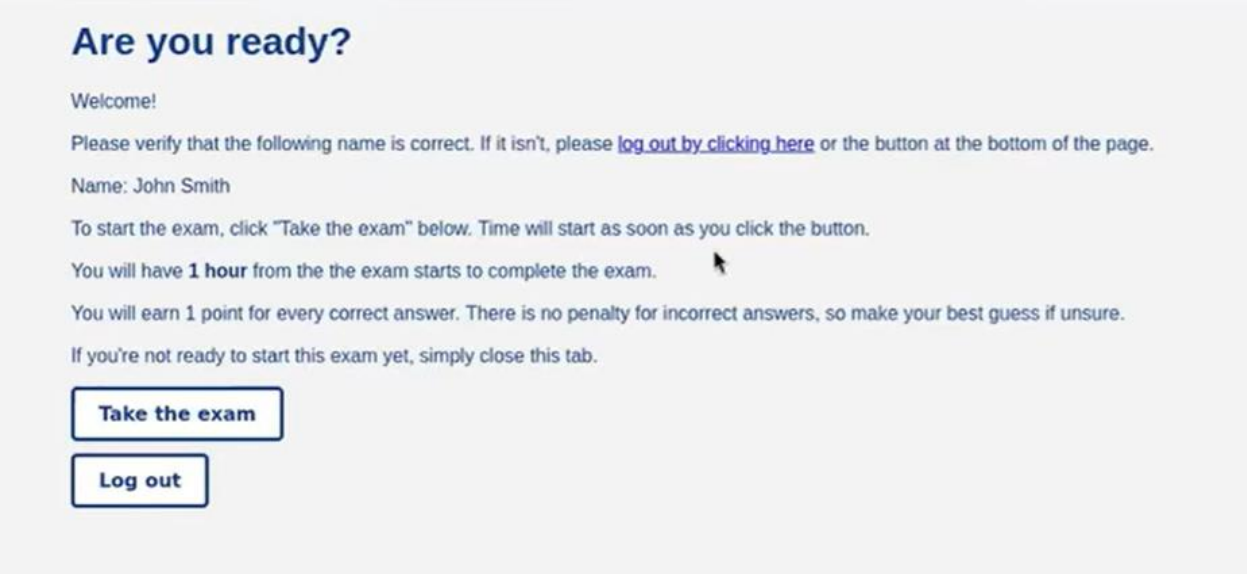


Enter your first and last name and the exam password your proctor provided, then click **Login**.



Verify that the following name is your name. If it isn't, click **Log out**.

If the name is correct, start the exam by clicking **Take the exam**.



You will have 1 hour to complete the exam. When you're finished answering the questions, click **Submit**.



The program will autograde your exam. If you score an 80% or higher, you will receive a certificate like the one below. If you receive this certificate, present it to clinic staff to move on to the onboarding process.



Registration

The general-purpose website is located in the **/website** subdirectory. All of the Python Flask code and HTML files for this part of the app are located within the **/webapp** subdirectory. There is a .env file that exists in the **webapp** directory that details all the environment variables for the website.

Environment variables:

MAIL_SERVER - The mail server used to send user verification emails

MAIL_PORT - Whether the mail server uses SMTP on port 25 or SMTPS on port 587

MAIL_USE_TLS - Tells the website to use TLS when sending emails

MAIL_USE_SSL - Tells the website to use SSL when sending emails

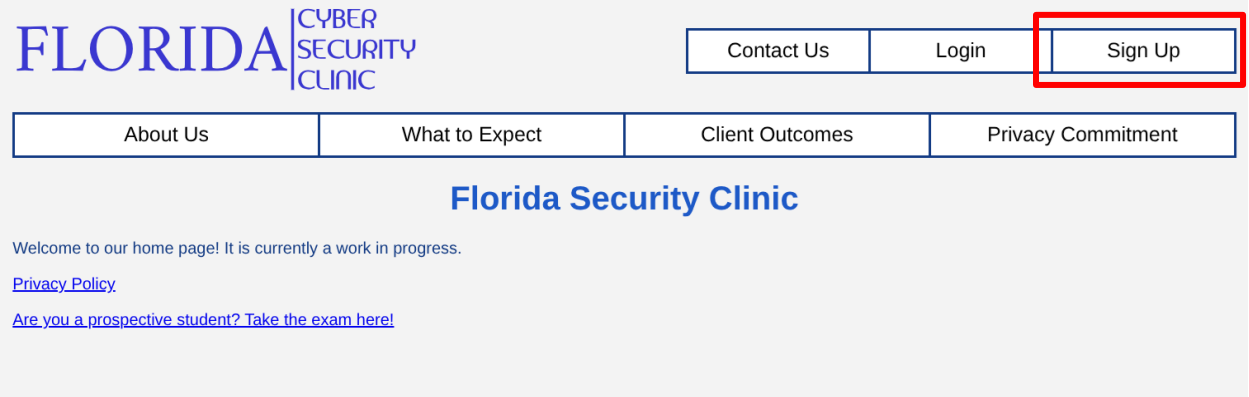
MAIL_USERNAME - The username for the mail server the website is configured to use.

MAIL_PASSWORD - The password used to authenticate to the mail server

MAIL_DEFAULT_SENDER - The address used to send the email

FLASK_SECRET_KEY - This is used to securely sign session cookies, as well as generate the CSRF token. This variable should be changed to a random long string in the .env file to ensure security.

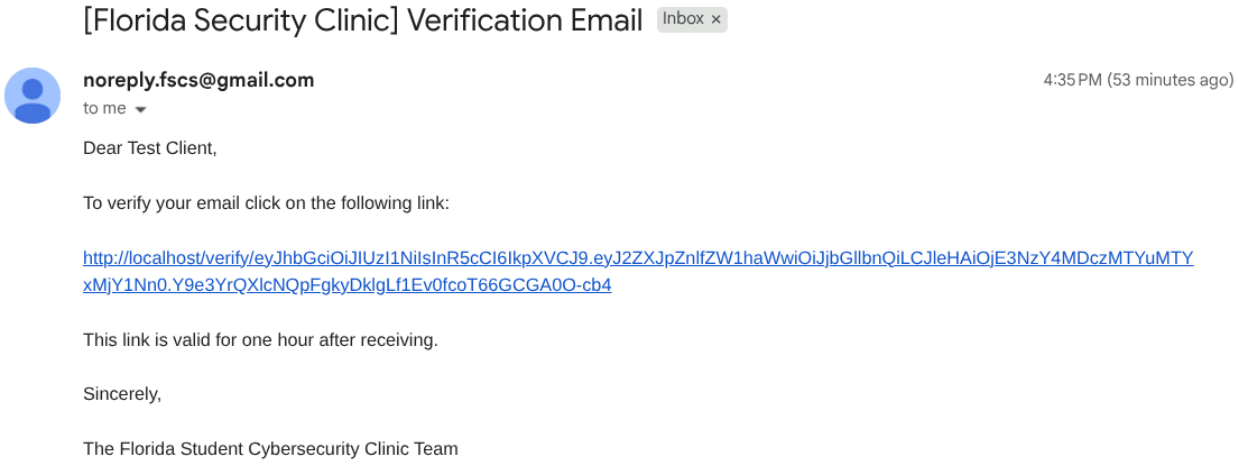
To create an account on the website, click the signup button pictured below.



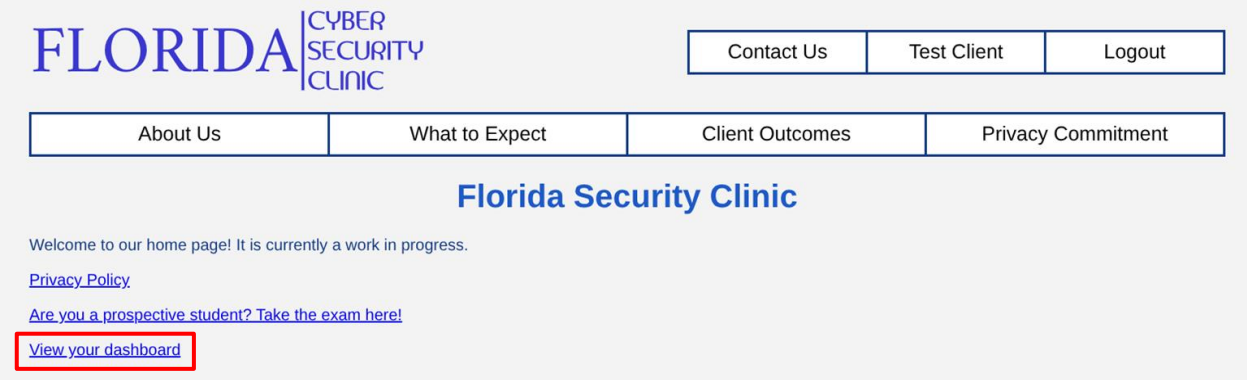
Then fill out and submit the form below. Choose student if you are creating a student login and client if you are creating a client login.



Once you submit the form, you will receive an email asking you to verify your email. This is required before you can login.



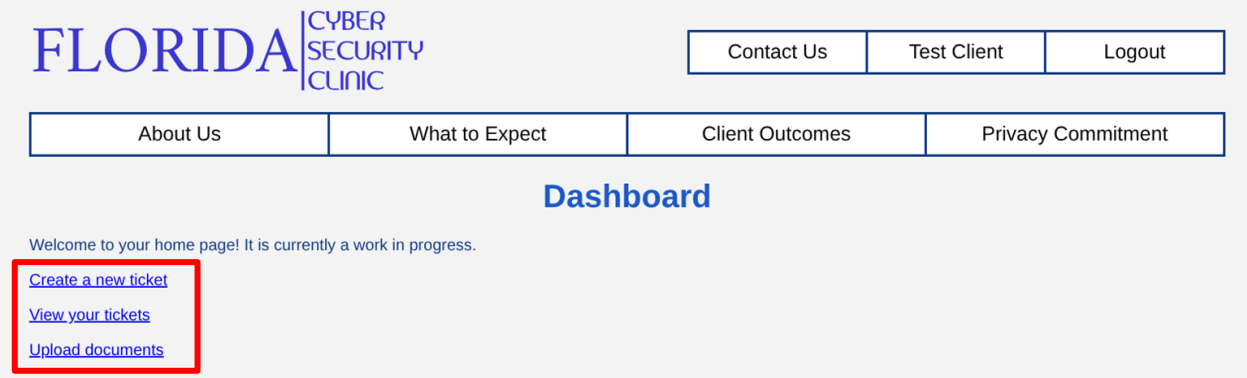
You now have access to your dashboard where you can view/create tickets or upload files.



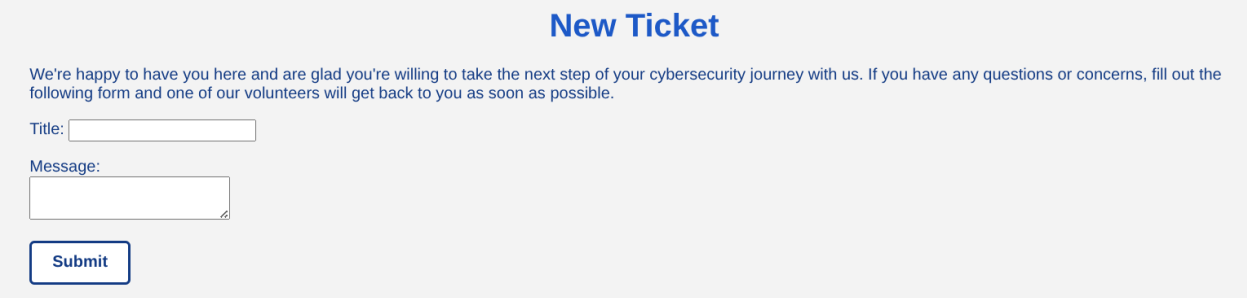
Ticketing System/File Upload

For Clients

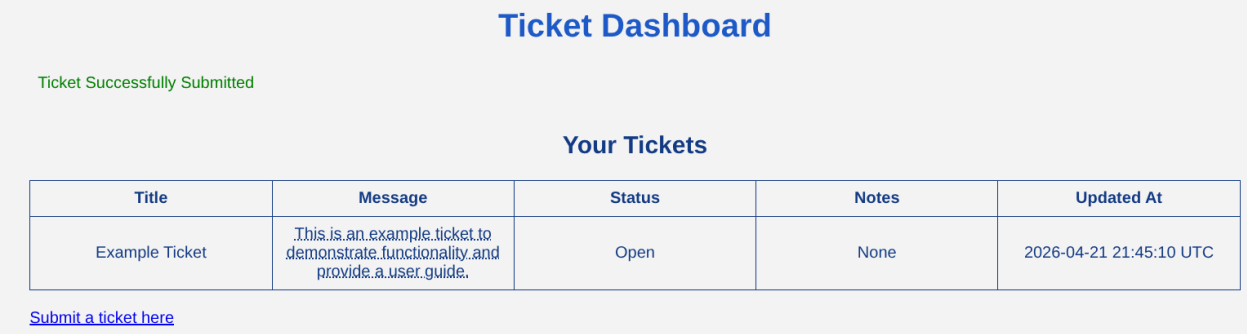
From your dashboard, you can create a new ticket, view your tickets, or upload a file.

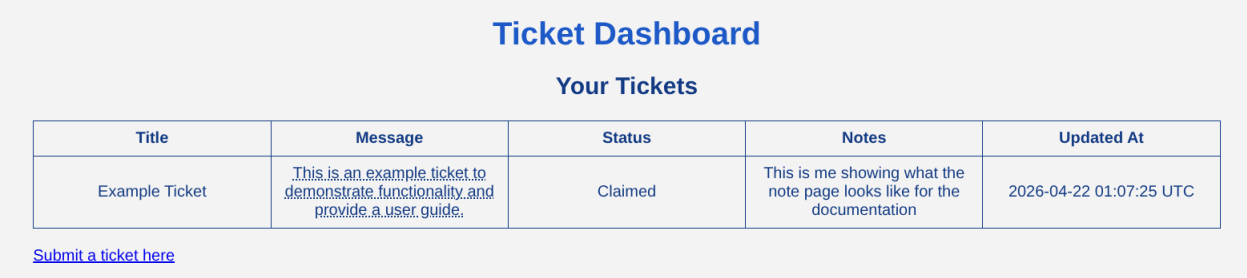


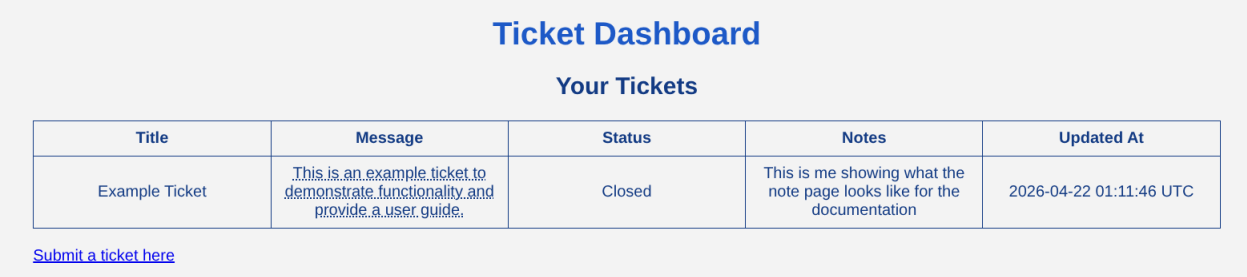
To create a new ticket, click on **Create a new ticket** and fill out the form that is pulled up.



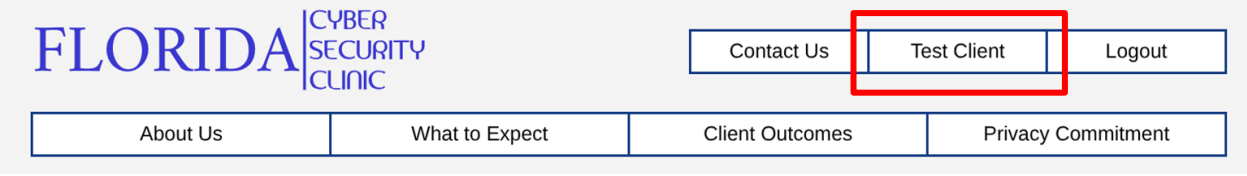
Once you submit your ticket, it redirects you to the ticket dashboard, where you can see all the tickets you've submitted. The tickets are sorted by the **Updated At** field.



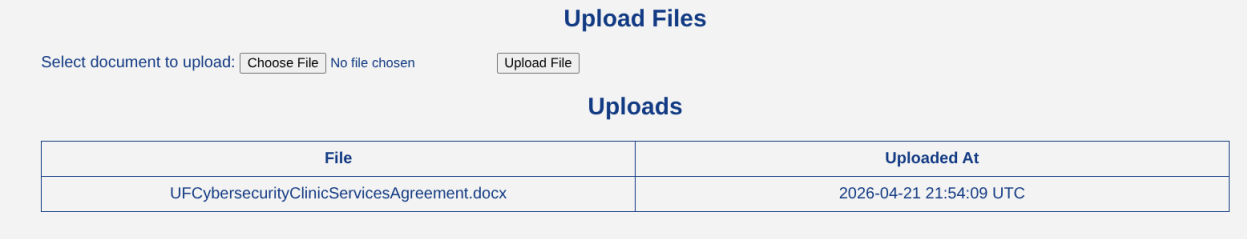




Whenever you want to go back to your dashboard, click on your as as shown below



To upload a file, click on **Upload documents** from your dashboard. Then attach your file using the form and submit it. You will see a table below listing the files you have uploaded to the clinic.

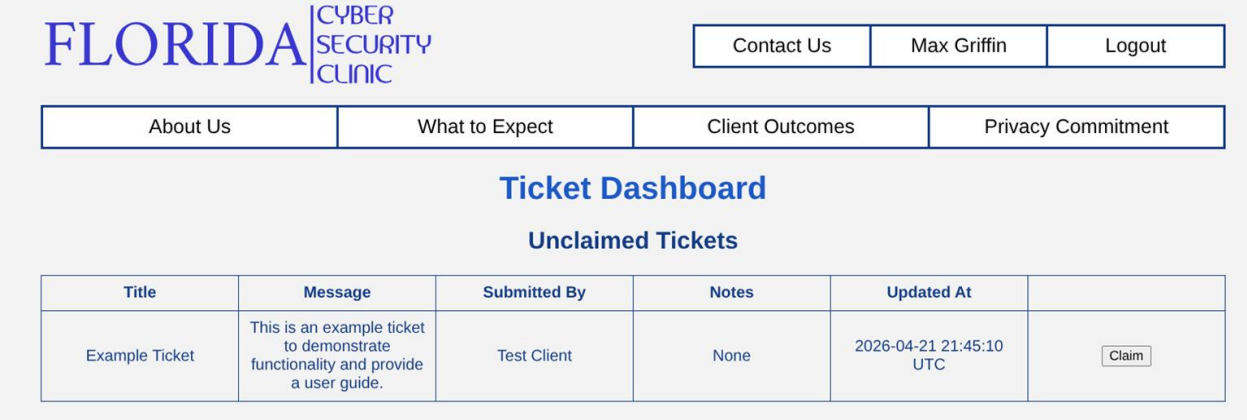


For Students

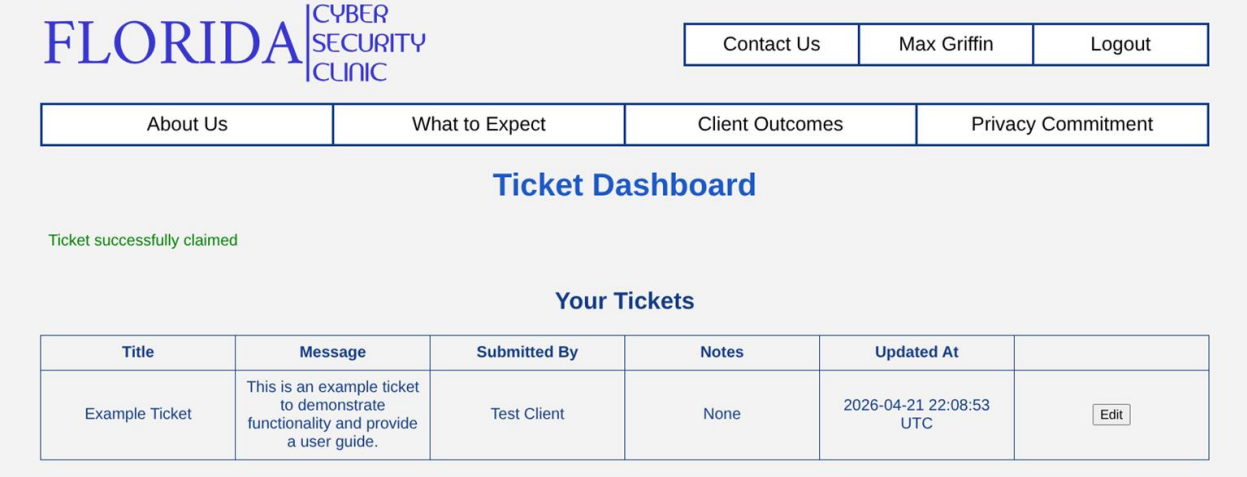
After logging in, click on **View your dashboard** on the home page.



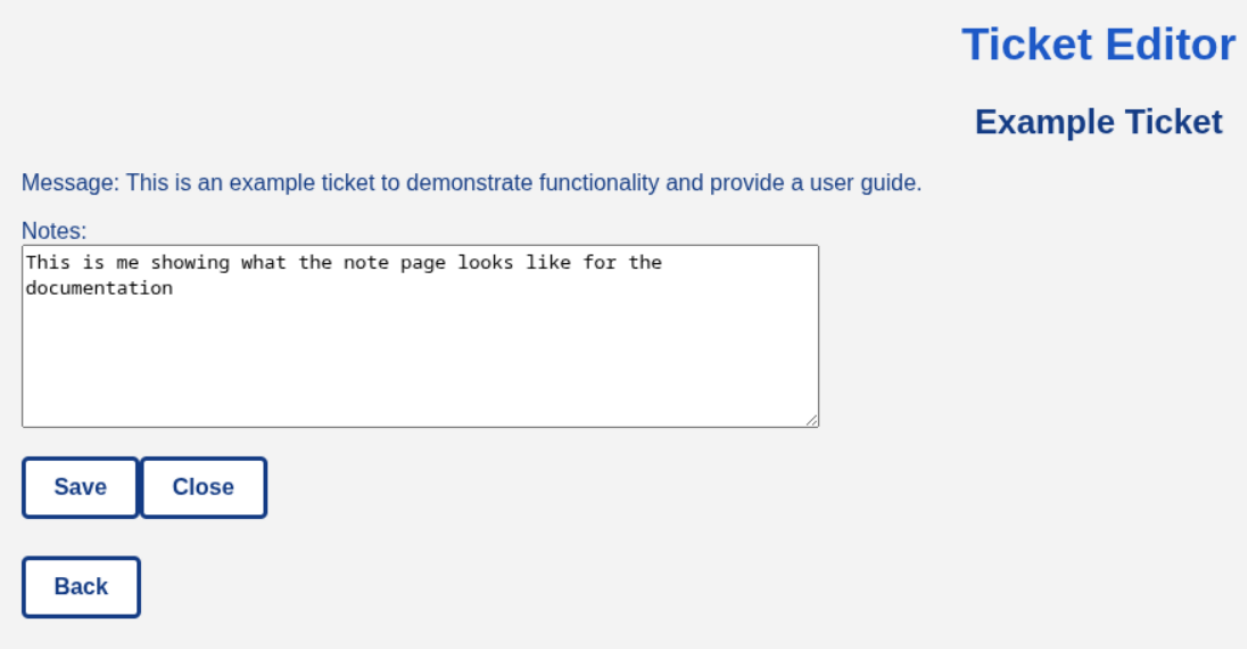
You have the same options as a client of **Create a new ticket**, **View your tickets**, and **Upload documents**. As a student, you will mainly use the **View your tickets** page. This is where you can claim, comment on, and close a client's ticket.



After claiming a ticket, you will have the option to edit the ticket. This is where you can add notes to the ticket, close the ticket, or both.



When you click on the edit button you are sent to a page to add notes to the ticket and then save it or close it. Clicking on **Save** will save the note, **Close** will save the note and close the ticket, and **Back** will discard the changes.



Transition Plan/Next Steps

Knowledge transfer

- Polish documentation.
- Write a process document for conducting a vulnerability assessment.
- Redact and archive data from engagement with [REDACTED].

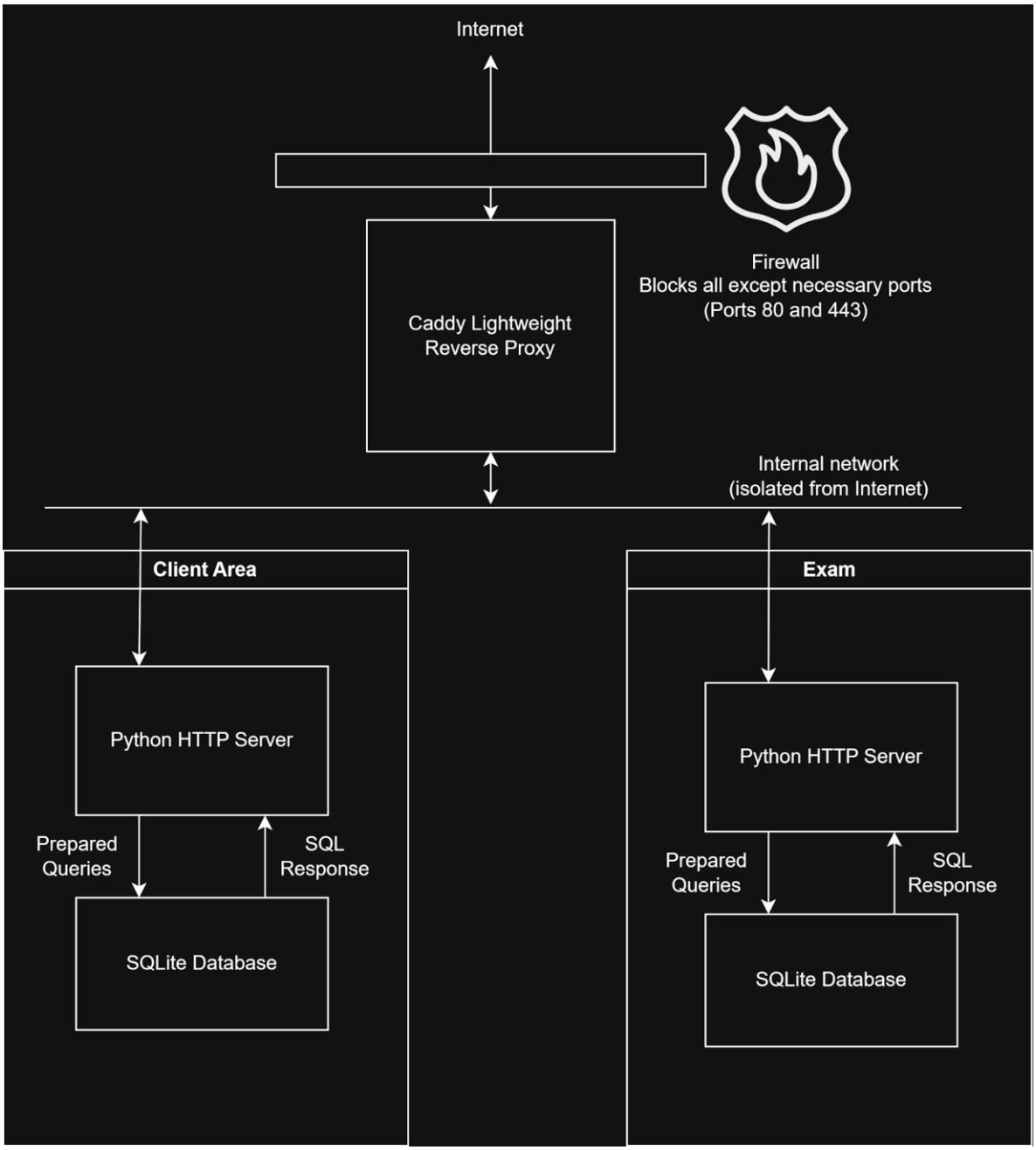
Sustainability

- Bringing FCSC to UF as a true student-run cybersecurity clinic.
- Build up trust in the clinic as we take on more satisfied clients.
- Oversight and education dispensed through a special topics (CIS4930) course where interested students can continue participating in the clinic.

Acknowledgments

We would like to thank [REDACTED] for the opportunity to work on a vulnerability assessment. We hope we were able to provide a valuable service to you. We also would like to thank Dr. Joseph Wilson for providing us the opportunity to work on this project, finding a client for us to work with, and guiding us throughout the process. We hope our work turns into a fruitful clinic that leads to a more secure society.

Appendix A


Internet
Firewall
Blocks all except necessary ports
(Ports 80 and 443)
Caddy Lightweight
Reverse Proxy
Internal network
(isolated from Internet)
Client Area
Exam
Python HTTP Server
Prepared
Queries
SQL
Response
SQLite Database
Python HTTP Server
Prepared
Queries
SQL
Response
SQLite Database

Biography

Adriel Barzola is hoping to break into the cybersecurity industry. He is interested in hopping back and forth between offensive and defensive security work.

Leo Graham is aiming to gain employment in the realm of cybersecurity or information technology. He is interested in threat analysis and the specificities of ongoing organized attack campaigns.

Maxwell Griffin is hoping to gain employment in a cybersecurity position as a SOC Analyst. He is interested in threat detection and response. Max would also like to someday work with developing or testing cryptographic algorithms.

Yuliang Huang is a computer science student at the University of Florida. Yuliang has done work for UFIT and the State of Florida in incident response and cyber defense. Yuliang plans to continue studying computer science in graduate school and break into the industry after that. Yuliang is personally interested in computer security and hopes that his future career will involve computer security in some way.

Sasha Watanabe has previously interned at CAMPUS USA Credit Union and Servco Pacific Inc. in cybersecurity roles. There, she has assisted with infrastructure projects, incident response plans, and security awareness programs. She is seeking roles that will involve her in the people side of security.